Kitchen Privacy Screen — Proposal

Cornelius, NC · November 10, 2026 · valid for 30 days · two options

Scope (both options)

- 22'-1" long, 8'-0" tall horizontal privacy screen with pergola top, finished on both sides, matching the reference image.
- **Posts:** pressure-treated pine 4x4 core (ground contact), set **24" deep** in concrete over 6" of gravel, holes dug with a rented mini skid steer and auger. Above grade, each post is clad in cedar: 2x6 faces and 1" side boards, all 5 1/2" wide with flush faces. The buried part is treated pine only, for lower cost and longer life in the soil.
- **Slats:** cedar 2x4 laid flat, held in notched side boards (even spacing, no visible fasteners on the slat faces).
- **Pergola top:** notched cedar 2x8 top boards on both sides, end caps for a solid-beam look, cedar 2x6 rafters with cloud-lift ends, one per post.
- **Fasteners:** coated trim-head deck screws (small countersunk heads, rated for treated lumber) and 1/2" galvanized through-bolts.
- All visible parts are pre-cut, notched and eased in our shop to keep on-site work short. Cleanup and debris removal included.

Comparison

	Option A — 10 posts	Option B — 7 posts
Posts	10	7
Post spacing (on center)	2'-4 13/16"	3'-7 1/4"
Clear bay (slat span)	23 5/16"	3'-1 3/4"
Warping risk	Low (< 36")	Moderate (> 36")
Slats	144	96
Materials	$4,275	$3,589
Labor (shop + site)	$4,747 (79 h)	$3,658 (61 h)
Service fee	$150	$150
Total	**$9,172**	**$7,398**

Option A uses the golden-ratio spacing with the shortest slat span (best against warping). Option B uses fewer posts and fewer parts (lower cost), with longer slats.

Materials and finish

Part	Material (species, grade, treatment)	Surface	Actual size	Qty A 10 posts	Qty B 7 posts
Post core (buried 24")	Pressure-treated Southern Yellow Pine 4x4, #2, ground contact (UC4B)	S4S (planed)	3 1/2" x 3 1/2" x 10'	10	7
Post faces (house + neighbor)	Western Red Cedar 2x6, kiln-dried	S4S (planed)	1 1/2" x 5 1/2" x 7'-5 1/2"	20	14
Notched side boards	Western Red Cedar 5/4x6, kiln-dried, ripped to 3 1/2"	S4S, eased edges	1" x 3 1/2" x 7'-5 1/2"	18	12
Plain side boards (end posts)	Western Red Cedar 5/4x6, kiln-dried, ripped to 3 1/2"	S4S, eased edges	1" x 3 1/2" x 7'-5 1/2"	2	2
Slats (laid flat)	Western Red Cedar 2x4, kiln-dried	S4S (planed)	1 1/2" x 3 1/2" x 2'-1 1/16" (A) / 3'-3 1/2" (B)	144	96
Top boards (notched for rafters)	Western Red Cedar 2x8, kiln-dried	S4S (planed)	1 1/2" x 7 1/4" (3 pieces per side, 8')	6	6
Top-board end caps	Western Red Cedar 2x8, kiln-dried	S4S (planed)	1 1/2" x 7 1/4" x 9 1/2"	2	2
Pergola rafters (cloud-lift ends)	Western Red Cedar 2x6, kiln-dried	S4S (planed)	1 1/2" x 5 1/2" x 36"	10	7

Finish: all cedar is Western Red Cedar, kiln-dried and **S4S (surfaced/planed on 4 sides)**, so every piece has true, consistent dimensions and the post faces, side boards and notches line up. Rough-sawn cedar (as stocked at some big-box stores in 2x6/2x8) is not used; if S4S is not stocked locally, it will be sourced from a lumber yard. Cedar is left natural (no stain needed) and will weather to silver-gray; sealer available on request.

Option A — 10 posts

- **10 composite posts** at 2'-4 13/16" on center, clear bay 23 5/16" (bay-to-post ratio 4.24 ≈ φ^3 (Fibonacci 55/13)). Slat span is well under the 36" limit we use to prevent warping.
- **144 cedar slats** (16 per bay, 1 11/16" gaps), 2'-1 1/16" long, held in notched side boards.
- **Pergola top:** 6 notched cedar top boards, 2 end caps, 10 rafters with cloud-lift ends.
- Drawings: sheets A-1 to A-3 at the end of this document.

Materials (incl. 8.25% Mecklenburg sales tax and equipment rental $450)	$4,275
Labor — shop pre-cutting 38 h + on-site installation 42 h (79 h × $60/h)	$4,747
Service fee	$150
Total Option A	**$9,172**

Included option: landscaping strip (landscape fabric, edging and river rock along the neighbor side), about $351 of the total; can be removed.

Option B — 7 posts

- **7 composite posts** at 3'-7 1/4" on center, clear bay 3'-1 3/4" (bay-to-post ratio 6.86 ≈ φ^4 (Fibonacci 89/13)). Slat span (3'-1 3/4") is above our conservative 36" limit (still under the 48" maximum for 2x4 cedar): slightly more risk of bowing over time than Option A.
- **96 cedar slats** (16 per bay, 1 11/16" gaps), 3'-3 1/2" long, held in notched side boards.
- **Pergola top:** 6 notched cedar top boards, 2 end caps, 7 rafters with cloud-lift ends.
- Drawings: sheets B-1 to B-3 at the end of this document.

Materials (incl. 8.25% Mecklenburg sales tax and equipment rental $450)	$3,589
Labor — shop pre-cutting 27 h + on-site installation 34 h (61 h × $60/h)	$3,658
Service fee	$150
Total Option B	**$7,398**

Included option: landscaping strip (landscape fabric, edging and river rock along the neighbor side), about $351 of the total; can be removed.

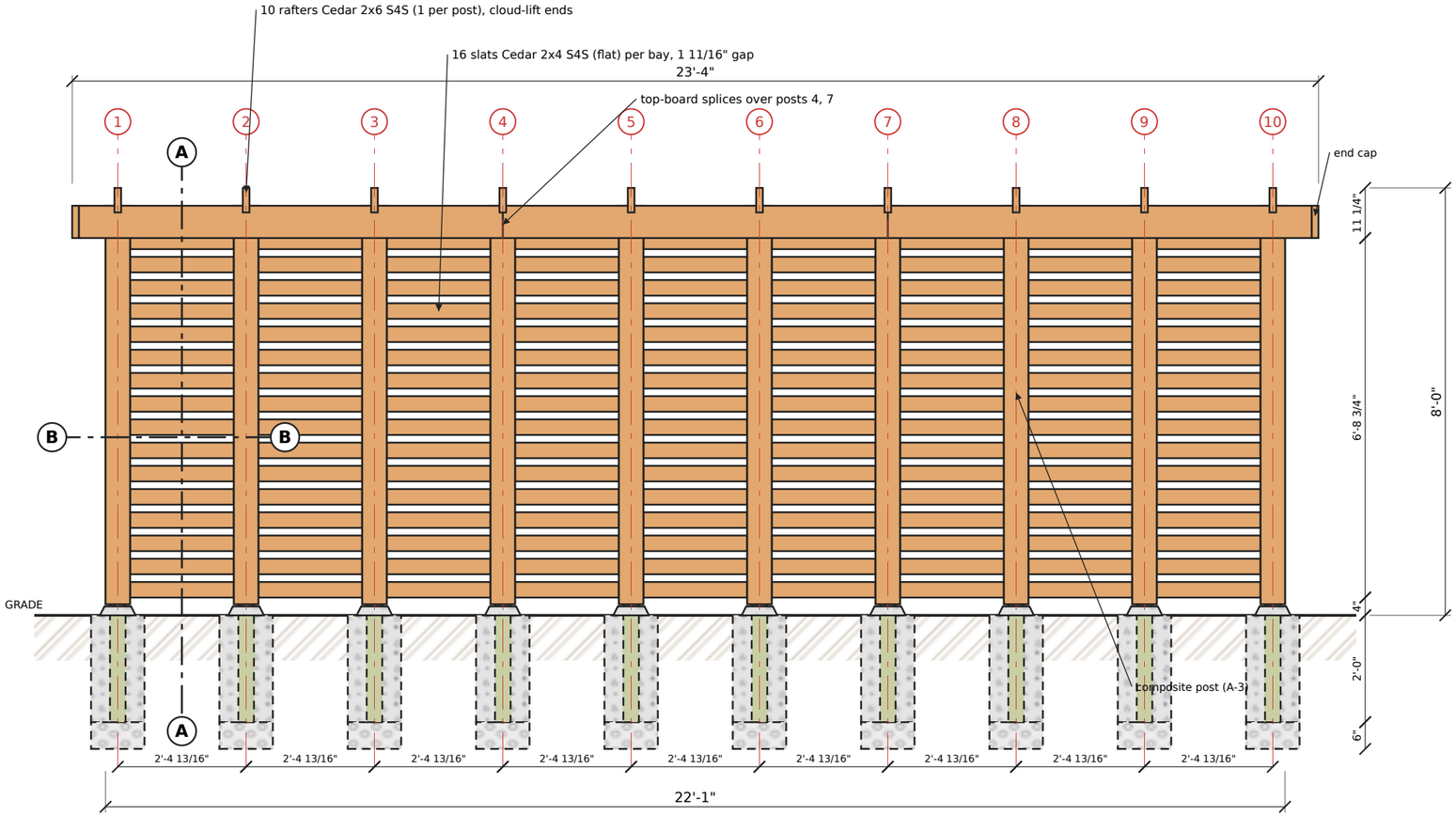
Installation sequence (both options)

1. Holes (machine auger)
2. Posts (treated pine cores in concrete)
3. Level post tops
4. Notched side boards (pre-cut), from the neighbor side
5. Slats (pre-cut)
6. Front closure (2x6 faces)
7. Side closure (end posts)
8. Top boards (pre-notched)
9. End caps (pre-cut)
10. Pergola rafters (pre-cut)

Notes, exclusions and payment

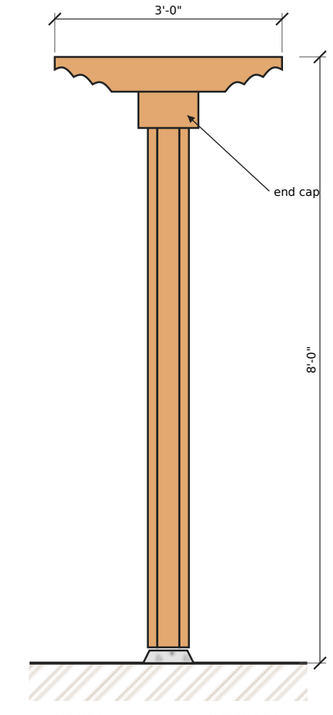
- Material prices are based on current Lowe's / Home Depot pricing and lumber-yard estimates; they will be confirmed at purchase. Changes above 5% may be adjusted.
- Not included: permits/HOA fees, surveys, rock or root removal beyond normal digging, plants/vines. We will call NC811 before digging.
- Payment: 40% deposit to order materials and start shop work · 40% when posts and slats are installed · 20% on completion.

Drawings follow: Option A (10 posts) sheets A-1 to A-3, then Option B (7 posts) sheets B-1 to B-3.



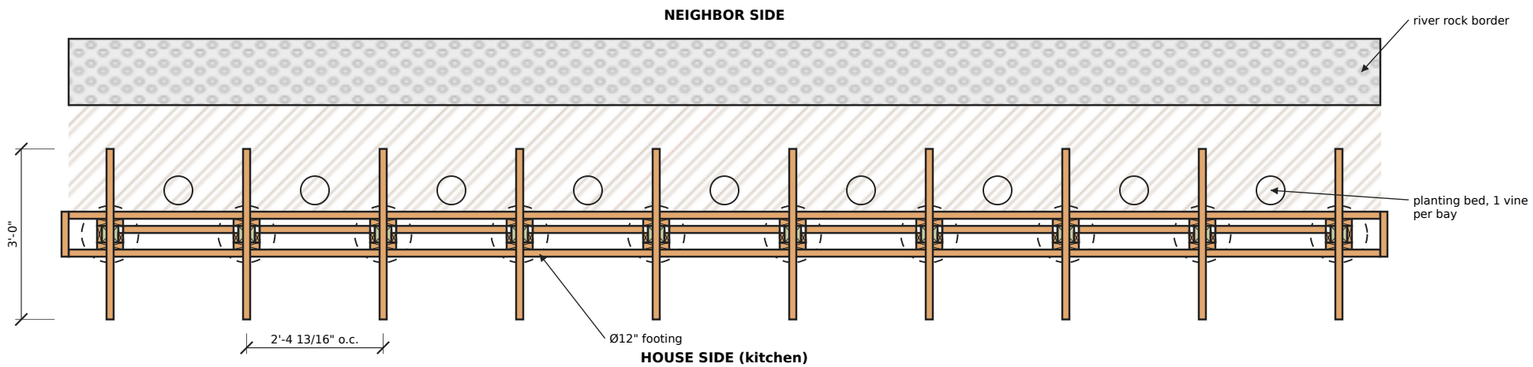
ELEVATION — NEIGHBOR SIDE

SCALE 1:32 (3/8" = 1'-0")



SIDE ELEVATION (end view)

SCALE 1:24 (1/2" = 1'-0")



PLAN — TOP VIEW

SCALE 1:32 (3/8" = 1'-0")

NOTES

1. Overall length 22'-1"; 10 posts at 2'-4 13/16" o.c.; clear bay 23 5/16".
2. Bay/post = 4.24 ≈ φ^3 (Fibonacci 55/13); span 23 5/16" < 36" (low warping risk).
3. Post cores embedded 2'-0" in concrete over 6" gravel; holes by machine auger.
4. Call NC811 before digging. Verify property line, setbacks and HOA.
5. Fasteners: coated trim-head deck screws (ACQ-rated), countersunk.
6. 16 slats per bay, 1 11/16" gap set by the notched side boards.
7. All posts 5 1/2" wide, faces flush. Parts marked YES are pre-cut in the shop.
8. Cedar = Western Red Cedar, kiln-dried, S4S (planed 4 sides).
9. PT = Southern Yellow Pine #2, ground contact.

Kitchen Privacy Screen — 10 posts, cedar with pergola top (Cornelius, NC)

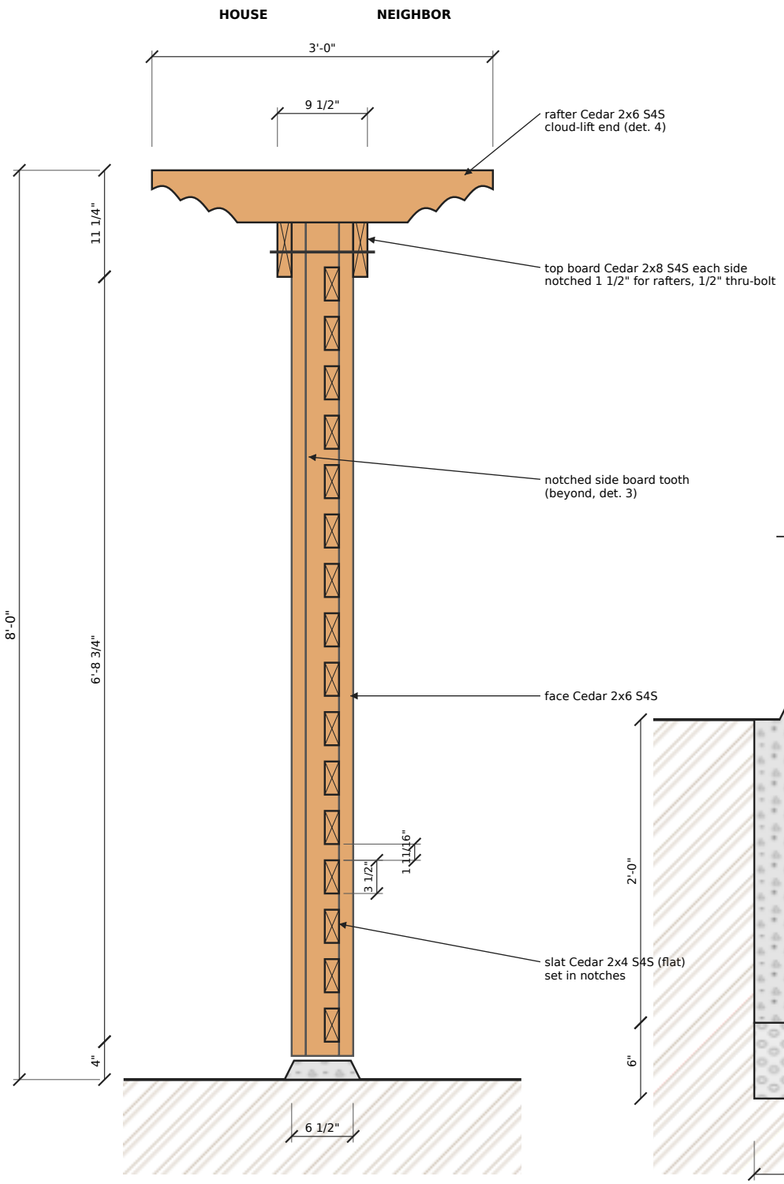
Elevation and plan

Date: 2026-11-10 Rev.: 7 By:

Dimensions in feet/inches. Lumber: actual sizes (2x6 = 1 1/2" x 5 1/2").

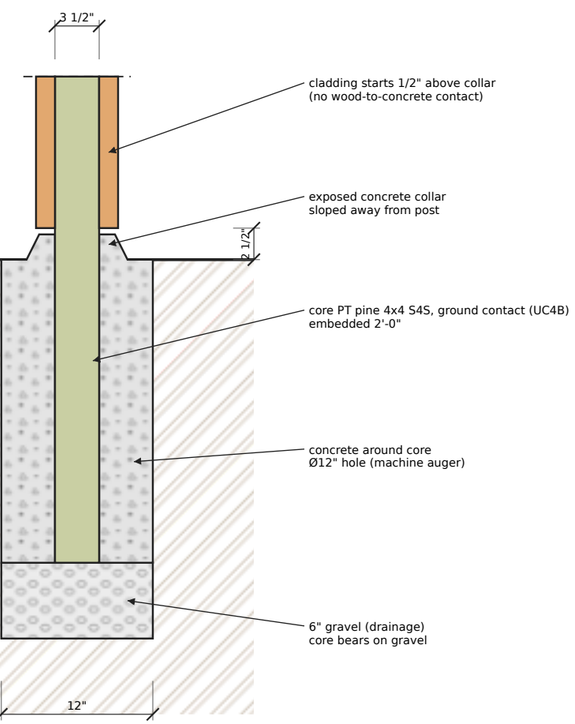
Verify all dimensions in the field before cutting. Not a structural design.

A-1



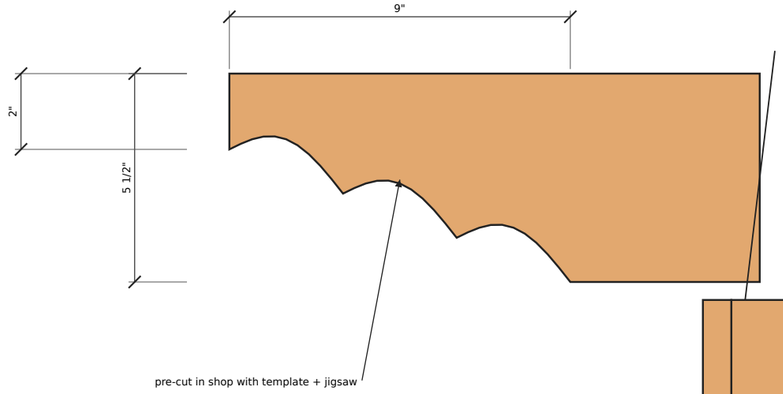
SECTION A-A (mid-bay, looking at post 1)

SCALE 1:16 (3/4" = 1'-0")



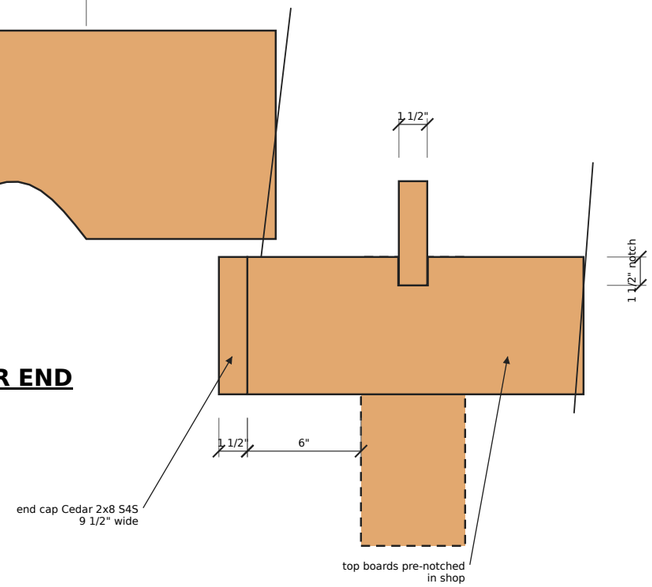
DETAIL 1 — POST FOOTING

SCALE 1:12 (1" = 1'-0")



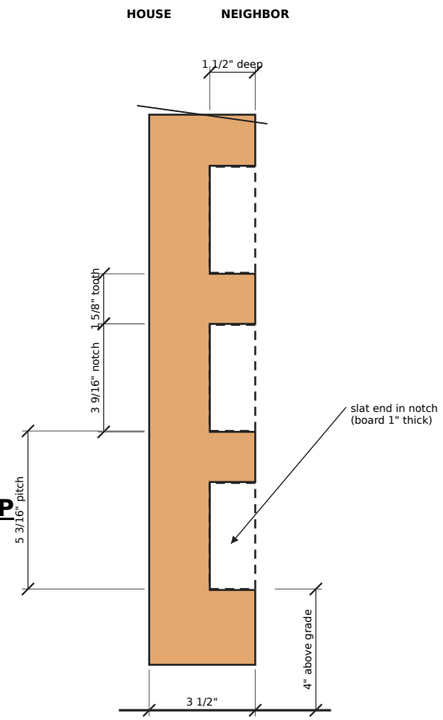
DETAIL 4 — RAFTER END

SCALE 1:4 (3" = 1'-0")



DETAIL 5 — TOP BOARD END: NOTCH + END CAP

SCALE 1:8 (1 1/2" = 1'-0")



DETAIL 3 — NOTCHED SIDE BOARD (from bay)

SCALE 1:5

PARTS LIST

Mark	Part	Material	Qty	Length	Shop pre-cut / notes
N	Post core	PT pine 4x4 S4S, ground contact (UC4B)	10	9'-8"	embed 2'-0"; trim tops on site
S	Notched side board	Cedar 5/4 S4S (1") notched	18	7'-5 1/2"	YES: rip to 3 1/2", 16 notches
R	Slat	Cedar 2x4 S4S (flat)	144	2'-1 1/16"	YES: cut to length (stop block)
F	Post face	Cedar 2x6 S4S	20	7'-5 1/2"	YES: cut to length
L	Plain side (end posts)	Cedar 5/4 S4S (1") plain	2	7'-5 1/2"	YES: rip to 3 1/2"
T	Top board (3 per side)	Cedar 2x8 S4S	6	7'-11 1/4", 7'-2 1/2"	YES: notches for rafters
E	Top board end cap	Cedar 2x8 S4S	2	9 1/2"	YES
K	Pergola rafter	Cedar 2x6 S4S	10	3'-0"	YES: cloud-lift ends

Kitchen Privacy Screen — 10 posts, cedar with pergola top (Cornelius, NC)

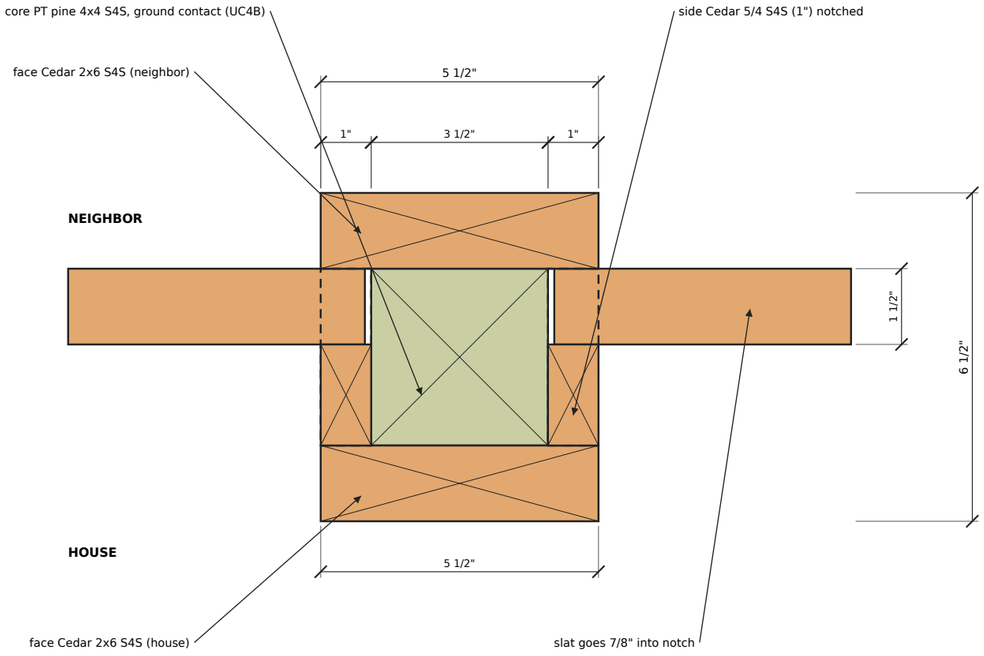
Sections and details

Date: 2026-11-10 Rev.: 7 By:

Dimensions in feet/inches. Lumber: actual sizes (2x6 = 1 1/2" x 5 1/2").

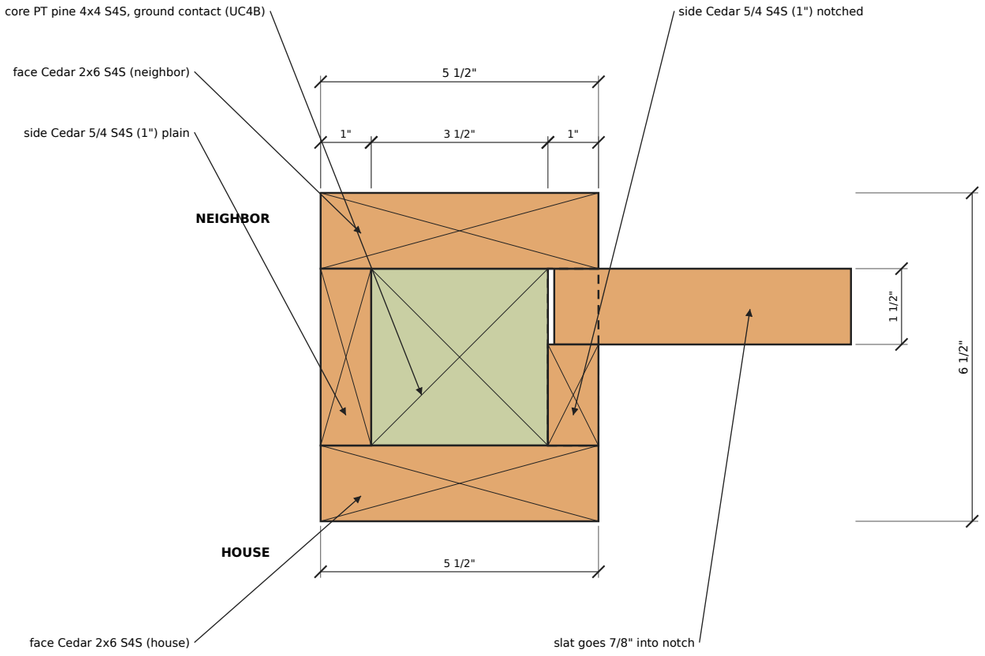
Verify all dimensions in the field before cutting. Not a structural design.

A-2



DETAIL 2a — INTERMEDIATE POST, SECTION B-B

SCALE 1:3



DETAIL 2b — END POST (1 and 10)

SCALE 1:3

#	INSTALLATION SEQUENCE
1	HOLES: lay out 10 post centers at 2'-4 13/16" o.c.; dig Ø12" x 2'-6" with rented machine auger; 6" gravel.
2	POSTS: set PT 4x4 cores 2'-0" deep, plumb and brace, pour concrete with sloped collar. Cure 48 h.
3	LEVEL TOPS: cut all cores to 7'-8" above grade (laser level).
4	NOTCHED SIDE BOARDS (pre-cut): fasten to core sides from the neighbor side, notches facing neighbor.
5	SLATS (pre-cut): drop into notches from the neighbor side, bottom to top; 1 screw per end.
6	FRONT CLOSURE: 2x6 face boards on both faces (closes the notches and locks the slats).
7	SIDE CLOSURE: 1" plain side board on the outer side of end posts 1 and 10.
8	TOP BOARDS (pre-notched): 2x8 each side of the posts, 1/2" thru-bolt per post; splices over posts 4, 7.
9	END CAPS (pre-cut): close the ends of the top boards for a solid-beam look.
10	PERGOLA RAFTERS (pre-cut): drop into top-board notches, 1 per post; hidden trim-head screws.

Kitchen Privacy Screen — 10 posts, cedar with pergola top (Cornelius, NC)

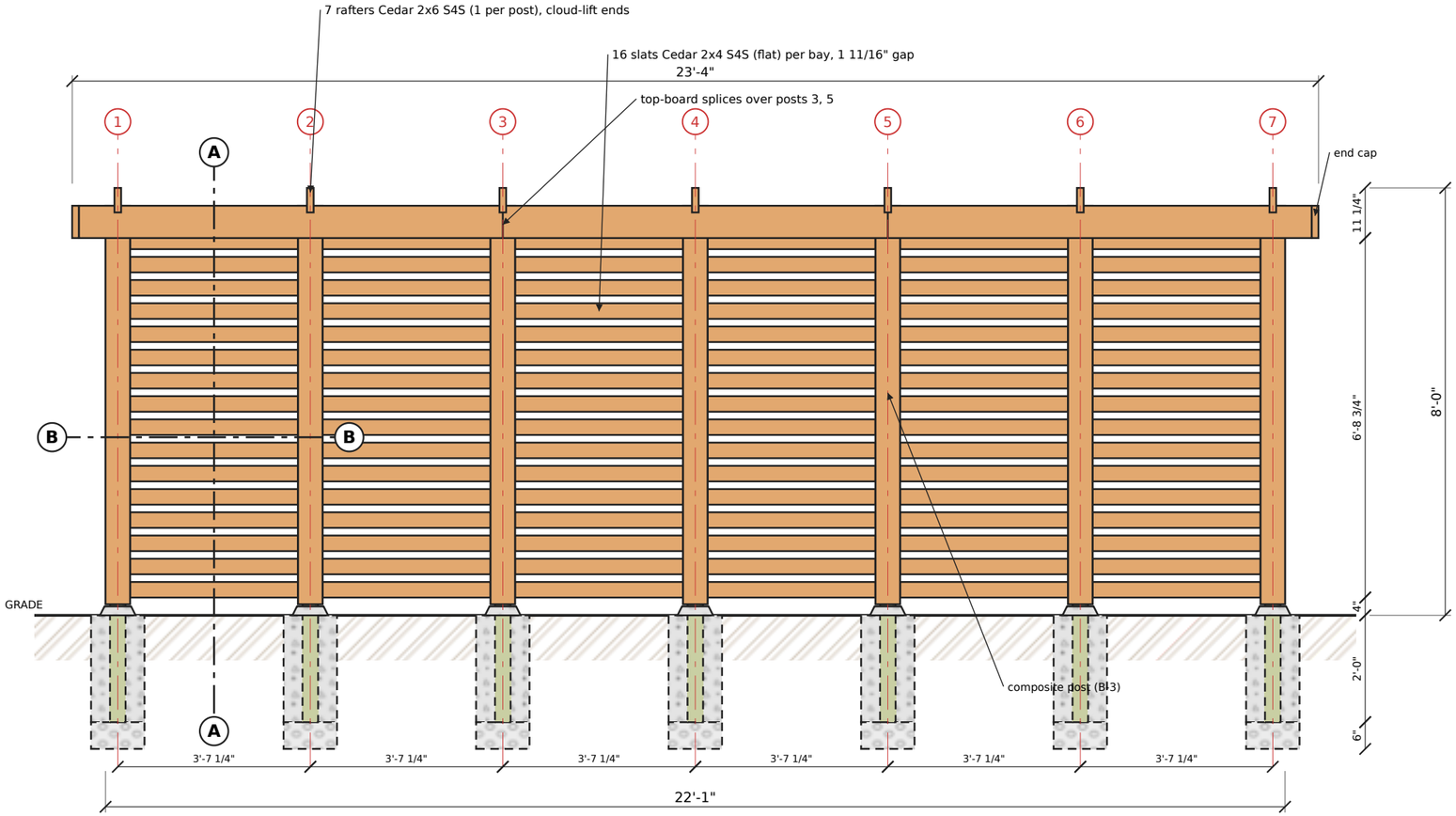
Composite post and installation

Date: 2026-11-10 Rev.: 7 By:

Dimensions in feet/inches. Lumber: actual sizes (2x6 = 1 1/2" x 5 1/2").

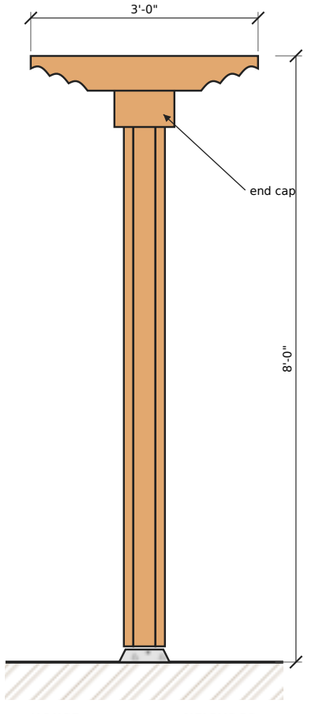
Verify all dimensions in the field before cutting. Not a structural design.

A-3



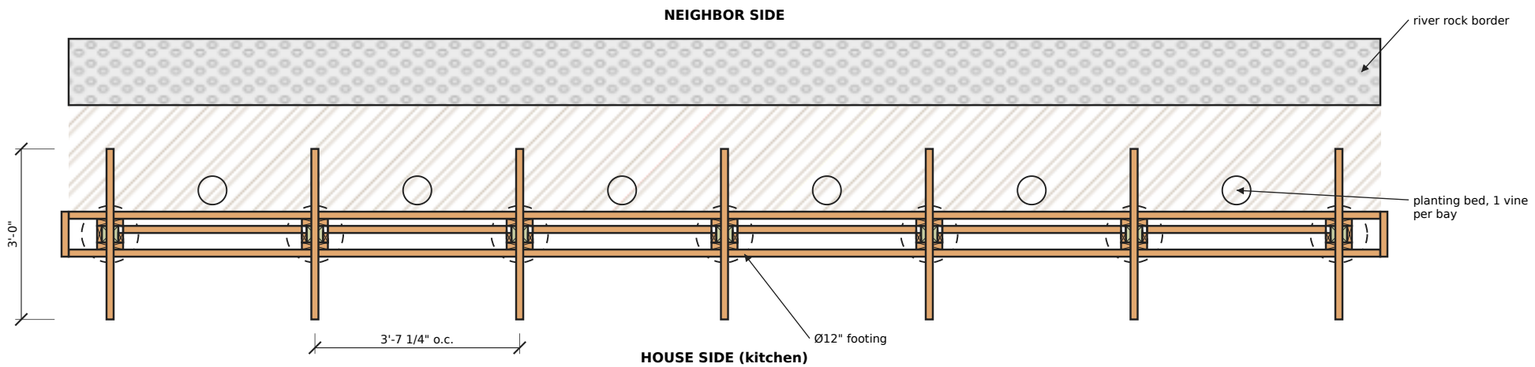
ELEVATION — NEIGHBOR SIDE

SCALE 1:32 (3/8" = 1'-0")



SIDE ELEVATION (end view)

SCALE 1:24 (1/2" = 1'-0")



PLAN — TOP VIEW

SCALE 1:32 (3/8" = 1'-0")

NOTES

1. Overall length 22'-1"; 7 posts at 3'-7 1/4" o.c.; clear bay 3'-1 3/4".
2. Bay/post = 6.86 ≈ φ^4 (Fibonacci 89/13); span 3'-1 3/4" > 36" limit (< 48" max).
3. Post cores embedded 2'-0" in concrete over 6" gravel; holes by machine auger.
4. Call NC811 before digging. Verify property line, setbacks and HOA.
5. Fasteners: coated trim-head deck screws (ACQ-rated), countersunk.
6. 16 slats per bay, 1 11/16" gap set by the notched side boards.
7. All posts 5 1/2" wide, faces flush. Parts marked YES are pre-cut in the shop.
8. Cedar = Western Red Cedar, kiln-dried, S4S (planed 4 sides).
9. PT = Southern Yellow Pine #2, ground contact.

Kitchen Privacy Screen — 7 posts, cedar with pergola top (Cornelius, NC)

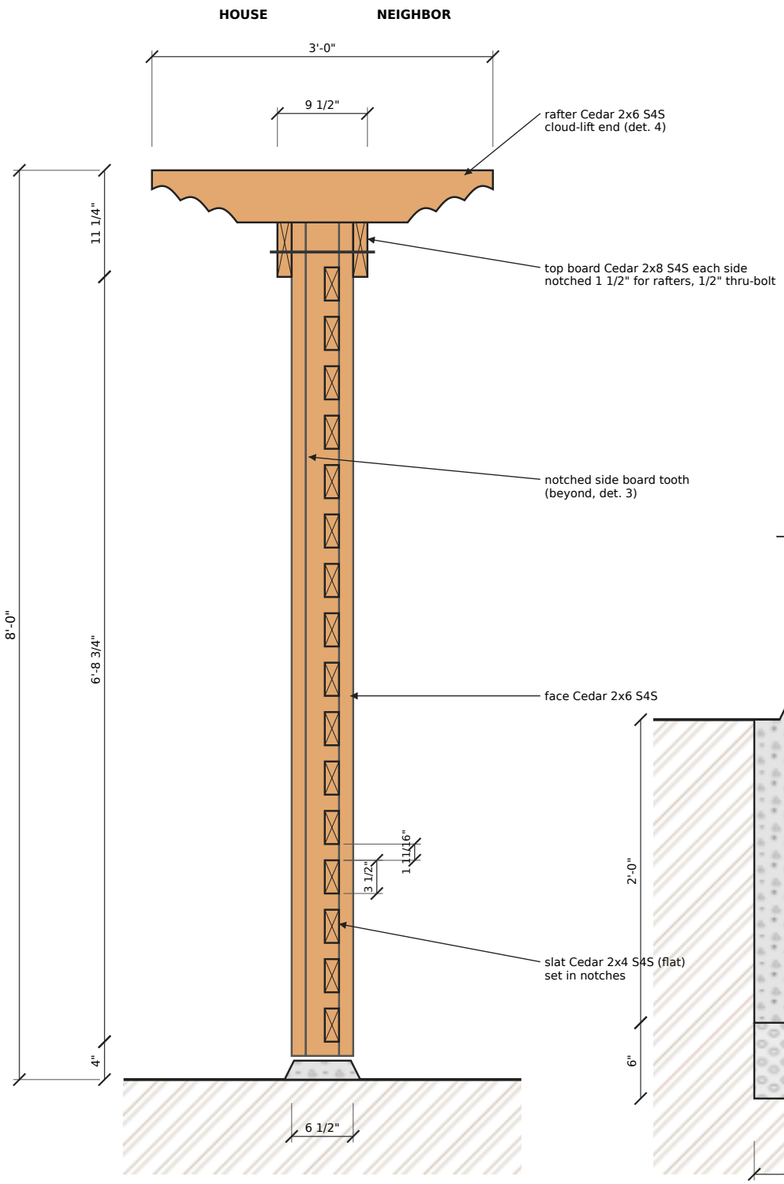
Elevation and plan

Date: 2026-11-10 Rev.: 7 By:

Dimensions in feet/inches. Lumber: actual sizes (2x6 = 1 1/2" x 5 1/2").

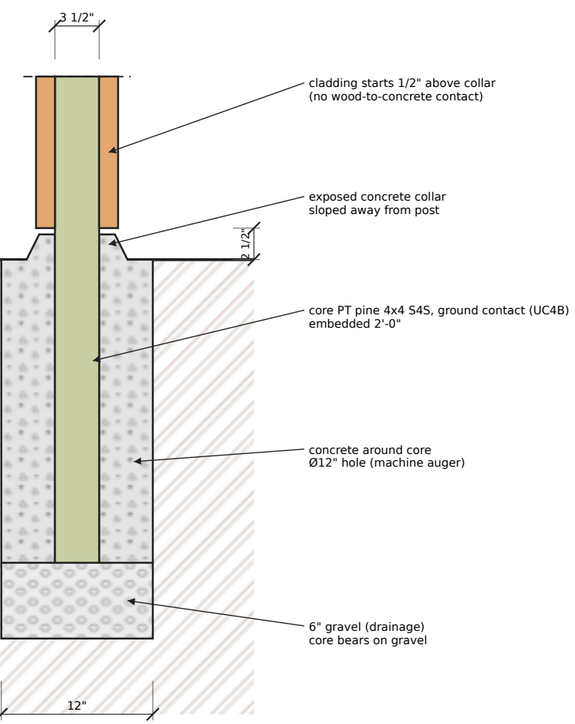
Verify all dimensions in the field before cutting. Not a structural design.

B-1



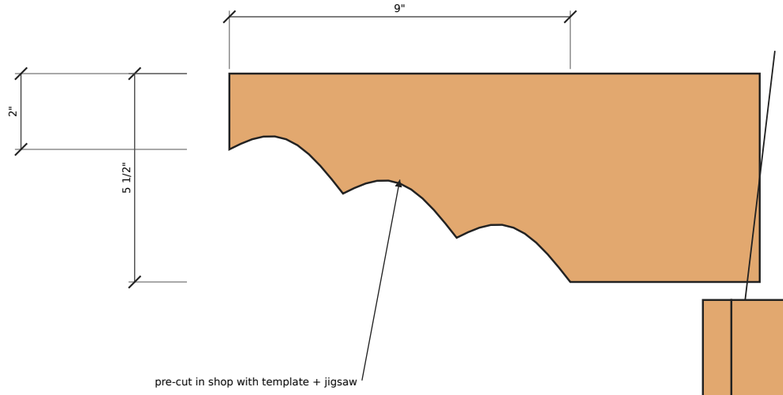
SECTION A-A (mid-bay, looking at post 1)

SCALE 1:16 (3/4" = 1'-0")



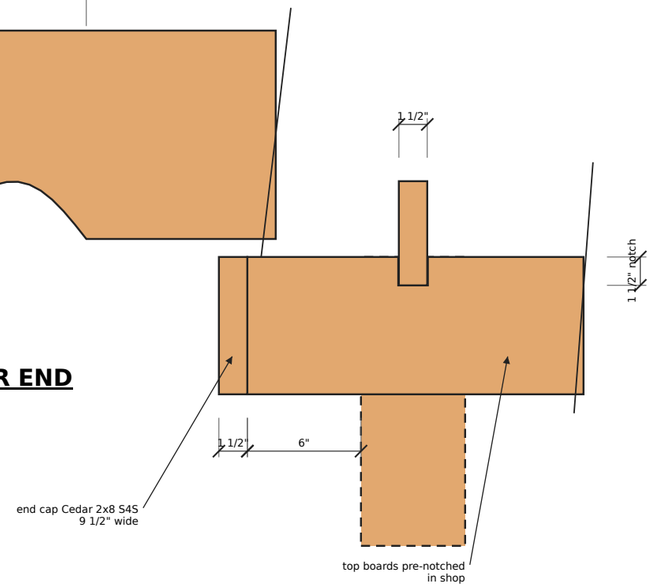
DETAIL 1 — POST FOOTING

SCALE 1:12 (1" = 1'-0")



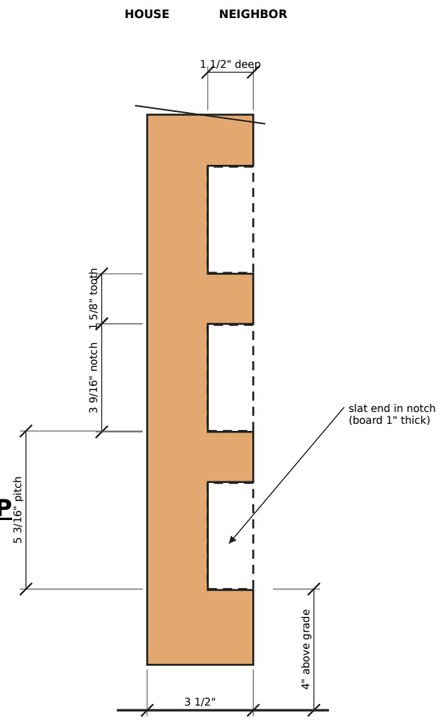
DETAIL 4 — RAFTER END

SCALE 1:4 (3" = 1'-0")



DETAIL 5 — TOP BOARD END: NOTCH + END CAP

SCALE 1:8 (1 1/2" = 1'-0")



DETAIL 3 — NOTCHED SIDE BOARD (from bay)

SCALE 1:5

PARTS LIST

Mark	Part	Material	Qty	Length	Shop pre-cut / notes
N	Post core	PT pine 4x4 S4S, ground contact (UC4B)	7	9'-8"	embed 2'-0"; trim tops on site
S	Notched side board	Cedar 5/4 S4S (1") notched	12	7'-5 1/2"	YES: rip to 3 1/2", 16 notches
R	Slat	Cedar 2x4 S4S (flat)	96	3'-3 1/2"	YES: cut to length (stop block)
F	Post face	Cedar 2x6 S4S	14	7'-5 1/2"	YES: cut to length
L	Plain side (end posts)	Cedar 5/4 S4S (1") plain	2	7'-5 1/2"	YES: rip to 3 1/2"
T	Top board (3 per side)	Cedar 2x8 S4S	6	7'-11 1/4", 7'-2 1/2"	YES: notches for rafters
E	Top board end cap	Cedar 2x8 S4S	2	9 1/2"	YES
K	Pergola rafter	Cedar 2x6 S4S	7	3'-0"	YES: cloud-lift ends

Kitchen Privacy Screen — 7 posts, cedar with pergola top (Cornelius, NC)

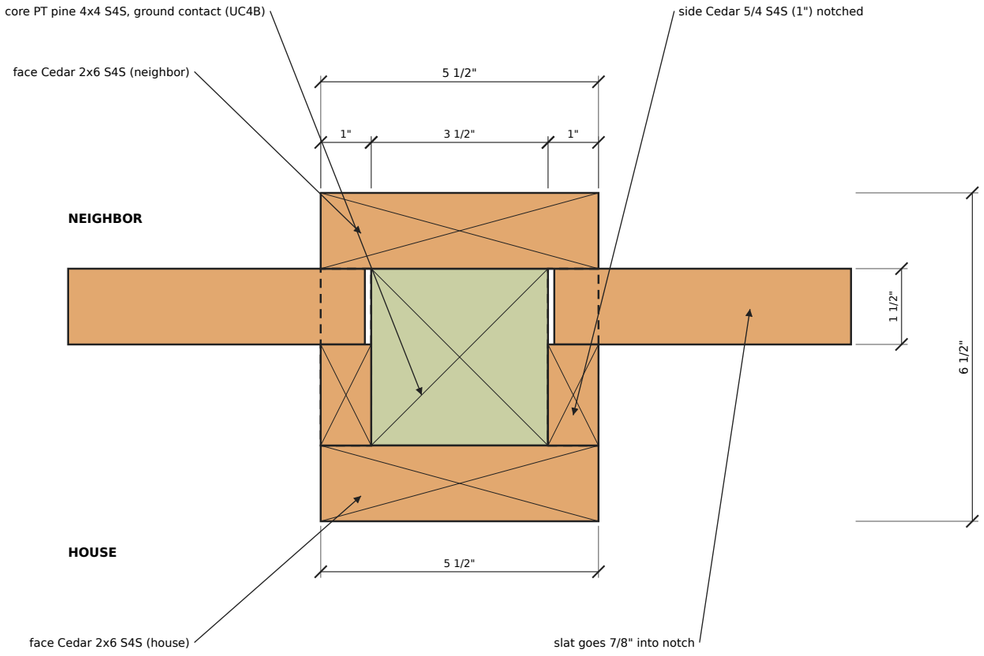
Sections and details

Date: 2026-11-10 Rev.: 7 By:

Dimensions in feet/inches. Lumber: actual sizes (2x6 = 1 1/2" x 5 1/2").

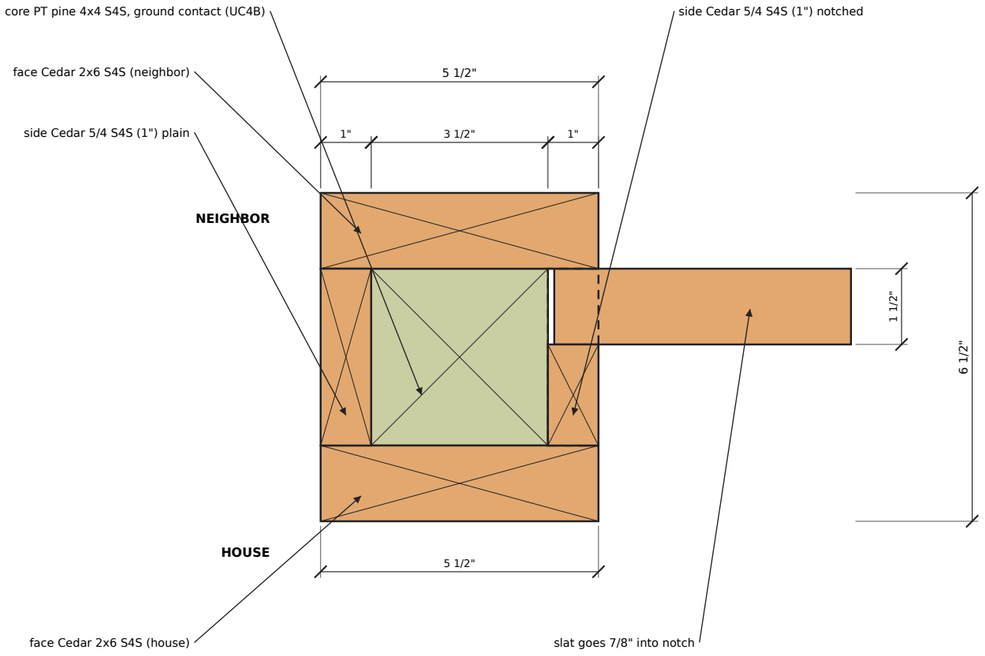
Verify all dimensions in the field before cutting. Not a structural design.

B-2



DETAIL 2a — INTERMEDIATE POST, SECTION B-B

SCALE 1:3



DETAIL 2b — END POST (1 and 7)

SCALE 1:3

#	INSTALLATION SEQUENCE
1	HOLES: lay out 7 post centers at 3'-7 1/4" o.c.; dig Ø12" x 2'-6" with rented machine auger; 6" gravel.
2	POSTS: set PT 4x4 cores 2'-0" deep, plumb and brace, pour concrete with sloped collar. Cure 48 h.
3	LEVEL TOPS: cut all cores to 7'-8" above grade (laser level).
4	NOTCHED SIDE BOARDS (pre-cut): fasten to core sides from the neighbor side, notches facing neighbor.
5	SLATS (pre-cut): drop into notches from the neighbor side, bottom to top; 1 screw per end.
6	FRONT CLOSURE: 2x6 face boards on both faces (closes the notches and locks the slats).
7	SIDE CLOSURE: 1" plain side board on the outer side of end posts 1 and 7.
8	TOP BOARDS (pre-notched): 2x8 each side of the posts, 1/2" thru-bolt per post; splices over posts 3, 5.
9	END CAPS (pre-cut): close the ends of the top boards for a solid-beam look.
10	PERGOLA RAFTERS (pre-cut): drop into top-board notches, 1 per post; hidden trim-head screws.

Kitchen Privacy Screen — 7 posts, cedar with pergola top (Cornelius, NC)

Composite post and installation

Date: 2026-11-10 Rev.: 7 By:

Dimensions in feet/inches. Lumber: actual sizes (2x6 = 1 1/2" x 5 1/2").

Verify all dimensions in the field before cutting. Not a structural design.

B-3